Managing Linux with the Embedded Perspective

Exercise 6

(Debugging with GDB and OpenOCD)

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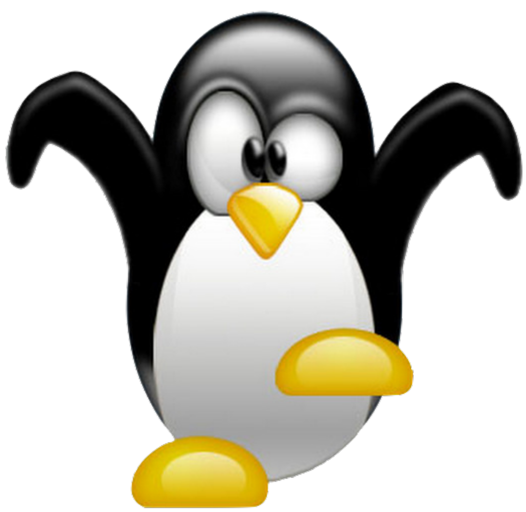

Introduction

In this exercise we will build upon the knowledge gained in the previous exercises, to build and deploy an embedded c program to a Raspberry Pi Pico with OpenOCD and debug it using GDB. The project that we will be using is the pio_encoder project that was downloaded and run in exercise 5

OpenOCD and GDB

You should use apt to install "gdb-multiarch" to be able to debug your raspberry pi pico. Please note that you may need to do some "Linuxing" to manage paths and such.

Some instructions on use can be found from the link below `https://www.raspberrypi.com/documentation/microcontrollers/debug-probe.html`

Build flash the project using CMake, make and OpenOCD so we can debug it. Document this step with a few screenshots.

Debugging

Once the program is running on your pico and the LED is blinking when you rotate the rotary encoder, you should start a GDB server in OpenOCD and connect to it from GDB.

You should now start to document your work (screenshots are helpful).

1. Show the source code in GDB.
2. You should experiment with breakpoints
3. You should experiment with watches
4. You should experiment with displaying expressions
5. You should experiment with stepping over and stepping into functions
6. You should experiment with backtrace (You will see more items on the stack after stepping into a few functions)
7. You should experiment with the info commands
8. You should experiment with watches

Please document with screenshots(with descriptions) and not any observations you have.